

Term Project Final

Overview of My Collected Data:

This visualization is a bubble plot that explores how my time management effectiveness affects or relates to my daily commitments. The main data I collected includes:

- Number of scheduled commitments per day.
- Total hours spent.
- Number of unexpected events.
- My self-rated effectiveness (1–5)

Below is how I collected my dataset.

Hypothesis:

As my daily commitment increases, my time management increases. On other factor that could impact my time management is unexpected engagements.

Date	Time (8-10pm)	Number of scheduled daily commitments	Number of unexpected engagements	Perceived time management effectiveness (1-5)	Notes
27/1/26	9pm	4	1	4 (Effective)	Busy but Productive
28/1/26	8:30pm	3	0	5 (Very Effective)	Very Productive
29/1/26	10pm	2	0	5 (Very Effective)	Very Productive
30/1/26	9pm	1	4	3 (Average)	Not so productive
31/1/26	9:40pm	2	6	1 (Very Poor)	Busy but not productiv
1/2/26	10pm	1	0	5 (Very Effective)	Very productive
2/2/26	9:20pm	3	5	1 (Very poor)	Busy but not productive
3/2/26	8:10pm	2	0	4 (Effective)	Productive

4/2/26	9pm	4	3	2 (Poor)	Not productive
5/2/26	9:15pm	2	0	4 (Effective)	Productive
6/2/26	8pm	3	1	3 (Average)	Not so productive
7/2/26	9:20pm	4	3	2 (Poor)	Not productive
8/2/26	10pm	2	5	3 (Average)	Busy but not productive
9/2/26	8:50pm	3	5	1 (Very poor)	Busy but not productive
10/2/26	9pm	2	0	5 (Very effective)	Busy but effective
11/2/26	8:40pm	4	5	1 (Very poor)	Very busy but not productive
12/2/26	8pm	4	0	5 (Very effective)	Busy and productive
13/2/26	9pm	3	6	1 (Very poor)	Busy but not productive
14/2/26	8:34pm	2	3	3 (Average)	Not so productive
15/2/26	9:10pm	2	0	4 (Effective)	Productive
16/2/26	9:40pm	4	0	5 (Very effective)	Productive
17/2/26	10pm	3	1	4 (Effective)	Productive
18/2/26	9:30pm	2	4	2 (Poor)	Not productive

Visualization Explanation:

I used a bubble chart to show the relationship between commitments and effectiveness.

- The x-axis shows the number of commitments

- The y-axis shows effectiveness (1–5)
- The size of the bubbles represents unexpected events
- The color of the bubbles represents hours worked (blue = fewer hours, red = more hours)

This format helps show multiple variables at once in a clear way.

Observations

Based on my visualization, I observed that higher numbers of unexpected events often correspond with lower effectiveness. Additionally, days with moderate commitments and fewer unexpected events tend to result in higher effectiveness. This suggests that both workload and unpredictability impact productivity. While the dataset is limited, it provides useful insights into how I manage time and respond to different daily conditions. I believe the amount of data collected was sufficient to identify general trends, although collecting more data over a longer period could improve the reliability of my conclusions. This suggests that having too many commitments or unexpected interruptions can reduce productivity.

Changes from Draft 2:

- I adjusted the bubble sizes to improve readability, clarity.

```
let radiusScale = d3.scaleSqrt()
```

```
.domain([0, 6])
```

```
.range([5, 50]); // reduced max size from 80 → 50
```

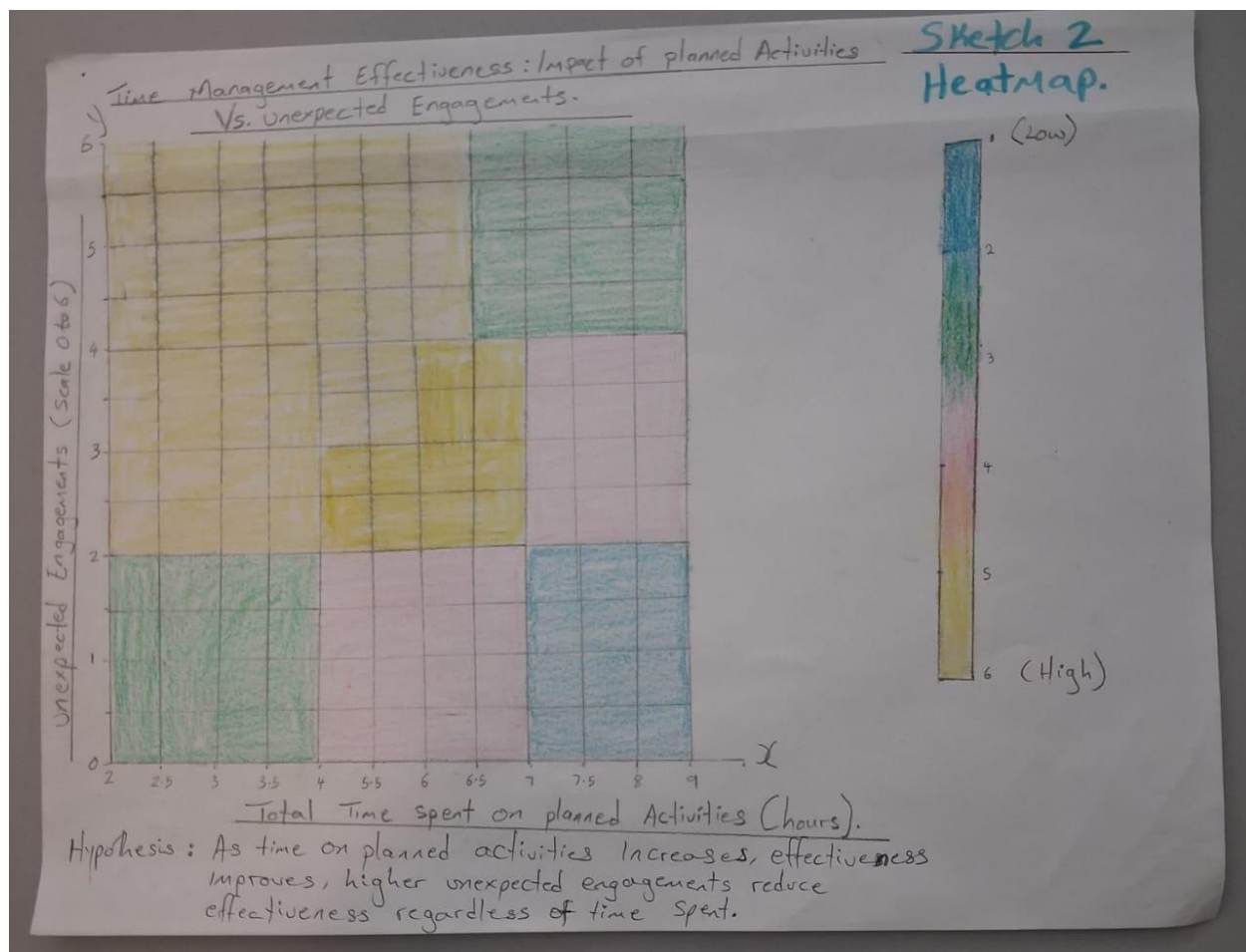
I adjusted the radius scale to reduce overlap and improve readability of the visualization.

- Keeps small circles the same
- Makes large circles less overwhelming
- Improves readability.

I also fixed issues in my code and improved my comments for better understanding. Converted arrow functions keys to regular functions key word.

ALTERNATIVE IDEA

I initially considered using a heatmap to visualize my data. However, I decided against this approach because heatmaps are better suited for showing patterns across two continuous variables or across time grids. My dataset focuses on relationships between multiple variables such as commitments, hours worked, unexpected events and effectiveness. A bubble plot was more appropriate because it allows me to represent multiple variables at once using position (x and y), size and color. This makes it easier to identify relationships between variables in a single view.



I didn't use heat map because my data has multiple variables and a bubble chart allowed me to show more information clearly in one view. A heat map is good for showing intensity or patterns across two variables, usually in a grid format. However, my dataset includes more than two important variables — commitments, effectiveness, hours, and unexpected events. It would not effectively represent the impact of unexpected events. The bubble plot was ultimately a better choice because it allows for more variables to be visualized simultaneously, making it more informative for my dataset.

Reflection

This project helped me better understand my daily habits and how different factors affect my productivity. I initially expected that more hours worked would always lead to higher effectiveness, but the data showed that unexpected events played a significant role. This project also improved my ability to work with data visualization tools such as D3 and strengthened my understanding of how to represent data clearly and effectively.

If I were to continue this project, I would collect additional data such as stress levels, sleep quality, or type of tasks completed. This could provide deeper insights into what factors most strongly influence

effectiveness. I would also explore adding interactivity to the visualization, such as tooltips or filtering options.